

Finite-Probe Inverse-Jet Reconstruction for Pandrosion

Symmetric probe stencils, tensor recovery, and derivative-free chart flattening

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Abstract

The all-order tensor inverse-jet theorem identifies the optimal local chart for a regular zero of an analytic system, but it is stated in terms of Taylor tensors. A derivative-free Pandrosion engine should recover those tensors from finite probes. This paper formulates and proves the local consistency statement behind that goal. For an analytic map $G : \mathbb{C}^n \rightarrow \mathbb{C}^n$ near a regular zero ρ , a fixed total-degree unisolvent probe cloud, evaluated at radius h , recovers the homogeneous Taylor coefficients G_q with error $\mathcal{O}(h)$. The reconstruction is done by one scaled polynomial fit, not by recursively subtracting lower-order $\mathcal{O}(h)$ approximations; this avoids the loss of powers of h that would otherwise contaminate higher tensors. Applying the inverse-jet recursion to the recovered tensors gives estimated chart coefficients $\hat{B}_q(h)$ satisfying

$$\hat{B}_q(h) \longrightarrow B_q, \quad 1 \leq q \leq m,$$

where B_q are the exact all-order inverse-jet coefficients. Consequently the reconstructed chart $\hat{\phi}_m$ is asymptotically m -flat:

$$G(\hat{\phi}_m(u)) = u + \mathcal{O}_h(\|u\|^{m+1}) + o(1)$$

on bounded chart coordinates. The paper closes with a finite-slope engine conjecture: stable Pandrosion probes with rank filtering and equal-value avoidance implement this reconstruction asymptotically without explicit derivative access.

Scope and status. The main results are local consistency theorems for finite probe reconstruction. They do not claim global convergence of a solver; they supply the derivative-free bridge between tensor inverse jets and practical Pandrosion chart engines.

1 Problem

The all-order inverse-jet chart at a regular zero is

$$\phi_m(u) = \rho + \sum_{q=1}^m B_q[u, \dots, u],$$

chosen so that

$$G(\phi_m(u)) = u + \mathcal{O}(\|u\|^{m+1}).$$

The coefficients B_q are computable from the Taylor tensors of G . A Pandrosion engine, however, observes values of G and finite slopes, not exact derivatives. We therefore ask:

Can the inverse-jet coefficients be reconstructed from finite probe evaluations as the probe radius tends to zero?

2 Probe stencils

Let H_q denote the finite-dimensional vector space of homogeneous polynomial maps $P : \mathbb{C}^n \rightarrow \mathbb{C}^n$ of degree q , and let

$$\Pi_m^0 = H_1 \oplus \cdots \oplus H_m$$

be the space of $\mathbb{C}^n$ -valued polynomials of total degree at most m with zero constant term.

Definition 2.1 (Homogeneous unisolvency). *A finite set of directions $V = \{v_1, \dots, v_N\} \subset \mathbb{C}^n$ is homogeneous-unisolvent through degree m if, for every $1 \leq q \leq m$, the evaluation map*

$$H_q \rightarrow (\mathbb{C}^n)^N, \quad P \mapsto (P(v_1), \dots, P(v_N))$$

is injective.

Definition 2.2 (Total-degree probe cloud). *A finite probe cloud $Y = \{y_1, \dots, y_M\} \subset \mathbb{C}^n$ is total-degree unisolvent through degree m if the evaluation map*

$$\Pi_m^0 \rightarrow (\mathbb{C}^n)^M, \quad P \mapsto (P(y_1), \dots, P(y_M))$$

is injective. Equivalently, a polynomial in Π_m^0 that vanishes on all points of Y must be zero.

Remark 2.3. *Generic probe clouds are total-degree unisolvent once the number of nodes is at least $\dim \Pi_m^0$. In implementation, one may use an overdetermined cloud and a fixed least-squares reconstruction operator. Homogeneous-unisolvent directions are still useful, but isolating the degree- q tensor from a single radius requires a total-degree fit or an equivalent multi-radius finite-difference formula.*

Write the Taylor expansion at a regular zero ρ as

$$G(\rho + w) = \sum_{q \geq 1} G_q[w, \dots, w], \quad G_1 = DG(\rho).$$

3 Homogeneous tensor recovery

Theorem 3.1 (Finite-probe Taylor tensor recovery). *Let G be analytic near ρ , and let Y be total-degree unisolvent through degree m . Let*

$$E_Y : \Pi_m^0 \rightarrow (\mathbb{C}^n)^M, \quad E_Y(P) = (P(y_1), \dots, P(y_M)),$$

and choose once and for all a bounded linear reconstruction operator $L_Y : (\mathbb{C}^n)^M \rightarrow \Pi_m^0$ such that $L_Y E_Y = \text{id}_{\Pi_m^0}$. For $h > 0$, define

$$\hat{P}_h(y) = \sum_{q=1}^m h^{q-1} \hat{G}_q(h)[y, \dots, y] = L_Y(h^{-1}G(\rho + hy_1), \dots, h^{-1}G(\rho + hy_M)).$$

When E_Y is square this is ordinary interpolation; when the cloud is overdetermined it is a fixed least-squares-type reconstruction. Then, for each $1 \leq q \leq m$,

$$\hat{G}_q(h) \rightarrow G_q \quad \text{as } h \rightarrow 0.$$

If the evaluations of G are exact, the error is $\mathcal{O}(h)$.

Proof. Taylor expansion gives, uniformly for y in the fixed finite cloud Y ,

$$h^{-1}G(\rho + hy) = \sum_{q=1}^m h^{q-1}G_q[y, \dots, y] + \mathcal{O}(h^m).$$

Let

$$P_h^*(y) = \sum_{q=1}^m h^{q-1}G_q[y, \dots, y] \in \Pi_m^0.$$

Thus the scaled data differ from $E_Y(P_h^*)$ by $\mathcal{O}(h^m)$. Applying the fixed bounded reconstruction operator L_Y gives $\hat{P}_h - P_h^* = \mathcal{O}(h^m)$ in coefficient norm. In particular, the degree- q homogeneous coefficient satisfies

$$h^{q-1}(\hat{G}_q(h) - G_q) = \mathcal{O}(h^m).$$

Hence $\hat{G}_q(h) - G_q = \mathcal{O}(h^{m-q+1})$, which is at least $\mathcal{O}(h)$ for $q \leq m$. This proves the claim. $\square$

Remark 3.2 (Why recursive subtraction is avoided). *If one first reconstructs $\hat{G}_\ell - G_\ell = \mathcal{O}(h)$ and then subtracts those lower orders before dividing by h^q , the lower-order errors are amplified by factors $h^{\ell-q}$. For $\ell < q - 1$, this amplification can diverge. The total-degree interpolation in [Theorem 3.1](#) isolates all degrees in one finite-dimensional solve, so no lower-order $\mathcal{O}(h)$ error is divided by a higher power of h .*

4 Inverse-jet reconstruction

The inverse-jet recursion is algebraic. Let $A = G_1$ and let B_q be the exact coefficients of the inverse jet. The first coefficient is $B_1 = A^{-1}$. At order k , the coefficient B_k solves

$$AB_k[u, \dots, u] + R_k(G_2, \dots, G_k; B_1, \dots, B_{k-1})(u) = 0,$$

where R_k is the homogeneous degree- k residual determined by lower data.

Theorem 4.1 (Consistency of finite-probe inverse jets). *Assume G_1 is invertible and the hypotheses of [Theorem 3.1](#) hold through degree m . Let $\hat{B}_q(h)$ be obtained by applying the inverse-jet recursion to $\hat{G}_1(h), \dots, \hat{G}_q(h)$. Then*

$$\hat{B}_q(h) \rightarrow B_q, \quad 1 \leq q \leq m.$$

With exact evaluations, $\hat{B}_q(h) - B_q = \mathcal{O}(h)$ for every fixed $q \leq m$.

Proof. The map $G_1 \mapsto G_1^{-1}$ is continuous near the invertible matrix G_1 , so $\hat{B}_1(h) \rightarrow B_1$. Suppose the result holds through order $k-1$. The residual polynomial R_k is a finite algebraic expression in $G_2, \dots, G_k$ and $B_1, \dots, B_{k-1}$; therefore its finite-probe estimate converges to the exact residual. Multiplication by $\hat{G}_1(h)^{-1}$ is also continuous. Thus $\hat{B}_k(h) \rightarrow B_k$. The $\mathcal{O}(h)$ rate follows from the same induction because all algebraic maps are locally Lipschitz on bounded finite-dimensional neighborhoods with G_1 bounded away from singularity. $\square$

Corollary 4.2 (Asymptotic m -flatness). *Let*

$$\hat{\phi}_m^{(h)}(u) = \rho + \sum_{q=1}^m \hat{B}_q(h)[u, \dots, u].$$

For every fixed bounded set of u -coordinates,

$$G(\hat{\phi}_m^{(h)}(u)) = u + \mathcal{O}(\|u\|^{m+1}) + \mathcal{O}(h) \sum_{q=1}^m \|u\|^q.$$

In particular, $\hat{\phi}_m^{(h)}$ converges coefficientwise to an m -flat inverse-jet chart.

Proof. Substitute the coefficient convergence from [Theorem 4.1](#) into the finite Taylor expansion of G . The exact inverse jet cancels homogeneous terms through degree m , and the coefficient error contributes the stated lower-order perturbation. $\square$

5 Noise and conditioning

Proposition 5.1 (Stable overdetermined reconstruction). *Suppose the total-degree reconstruction operator has condition number at most κ_Y , and suppose probe evaluations have additive error at most ε_h . Then the recovered q -th Taylor tensor has an additional error bounded by*

$$\mathcal{O}(\kappa_Y \varepsilon_h h^{-q})$$

at order q . Consequently consistent order- m reconstruction requires $\varepsilon_h = o(h^m)$ for the highest tensor.

Proof. The scaled data $h^{-1}G(\rho + hy_j)$ inherit additive error at most $\varepsilon_h h^{-1}$. The fixed polynomial reconstruction amplifies this by at most κ_Y . The degree- q coefficient of $\hat{P}_h$ is $h^{q-1}\hat{G}_q(h)$, so division by h^{q-1} gives an additional tensor error of size $\mathcal{O}(\kappa_Y \varepsilon_h h^{-q})$. The highest recovered degree imposes the strongest scaling requirement. $\square$

6 Pandrosion engine conjecture

Conjecture 6.1 (Derivative-free inverse-jet reconstruction by stable probes). *A Pandrosion engine using symmetric total-degree probe clouds, finite-slope rank filtering, equal-value pole avoidance, and shrinking probe radius h reconstructs the all-order inverse-jet coefficients through degree m whenever the probe cloud remains uniformly unisolvent and the effective evaluation noise is $o(h^m)$.*

Conjecture 6.2 (Reconstructed chart multiplier law). *Under the hypotheses above, the charted anchored Pandrosion map built from the reconstructed chart satisfies*

$$\left\| DP_{G \circ \hat{\phi}_m^{(h)}, a}(0) \right\| \leq C \|a\|^m + o(1)$$

as $h \rightarrow 0$, uniformly for anchors in a fixed small chart ball.

7 Conclusion

The all-order inverse jet is not merely a formal derivative object. It can be recovered from finite probes, order by order, provided the stencil is rich enough and the probe radius is controlled. This makes the m -flat atlas programme algorithmic: exact derivatives define the target, while symmetric finite probes supply a derivative-free path toward it.

References

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